

## Laboratorio #5.1:

### Laboratorio 1:

#### Objetivos:

- Crear un Dockerfile para una aplicación Flask
- Construir una imagen Docker personalizada
- Publicar la imagen en Docker Hub
- Ejecutar el contenedor localmente
- Comparar peso de imagen si cambiamos la imagen base

#### Guías de laboratorio:

- [https://github.com/utec-devops-2026/laboratorios/blob/main/laboratorio\\_docker/laboratorio\\_docker\\_01.md](https://github.com/utec-devops-2026/laboratorios/blob/main/laboratorio_docker/laboratorio_docker_01.md)

### Entregables

1. **URL de repositorio** (GitHub o GitLab).
  - o <https://github.com/aarontimana04/lab-docker-flask.git>
2. Subir su repositorio usando buenas prácticas de git (GitFlow/TBD, conventional commits, PR)

Estructura del repo:

Ejemplo:

```
flask-docker-app/  
├─ app.py  
├─ requirements.txt  
├─ Dockerfile  
└─ .dockerignore
```

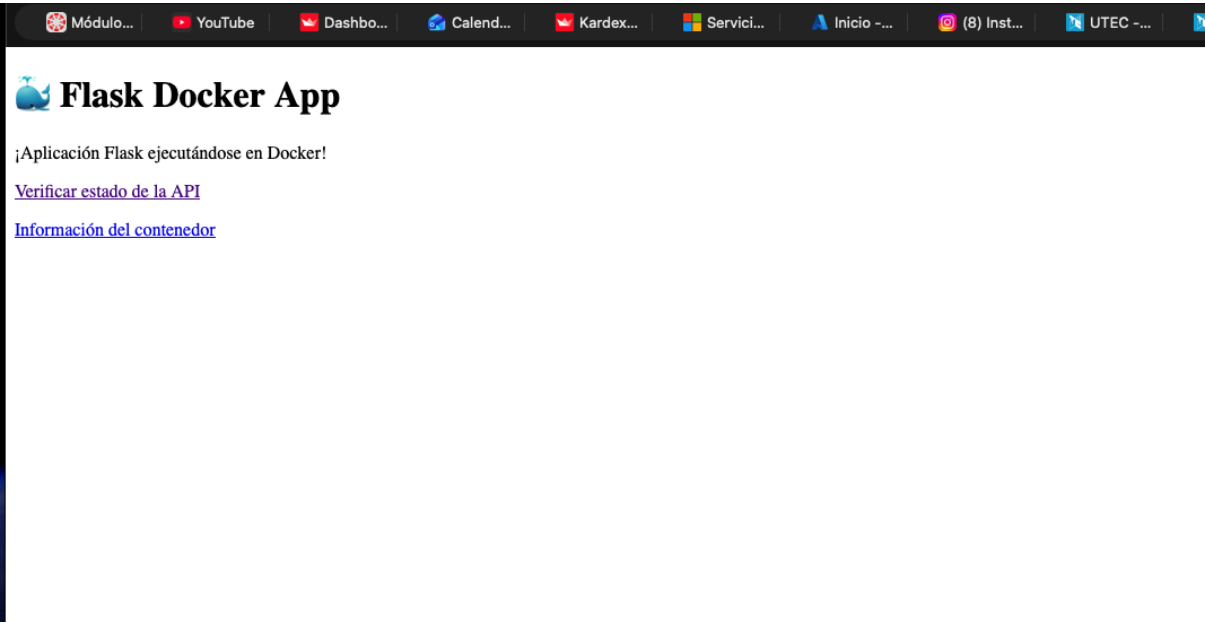
3. Captura de su imagen docker

|                          |                                  |                    |        |              |               |     |
|--------------------------|----------------------------------|--------------------|--------|--------------|---------------|-----|
| <input type="checkbox"/> | <input type="radio"/>            | cassandra          | latest | 5d5f627eb21e | 4 months ago  | 56  |
| <input type="checkbox"/> | <input checked="" type="radio"/> | mariadb            | 11.4.7 | 39596f079862 | 1 year ago    | 477 |
| <input type="checkbox"/> | <input checked="" type="radio"/> | mcr.microsoft.com/ | latest | 902628a8be89 | 4 years ago   | 2   |
| <input type="checkbox"/> | <input type="radio"/>            | nuvo/docker-cqlsh  | latest | 410c040df719 | 8 years ago   | 111 |
| <input type="checkbox"/> | <input checked="" type="radio"/> | flask-docker-app   | 1.0    | abb66c23fbd7 | 2 minutes ago | 249 |

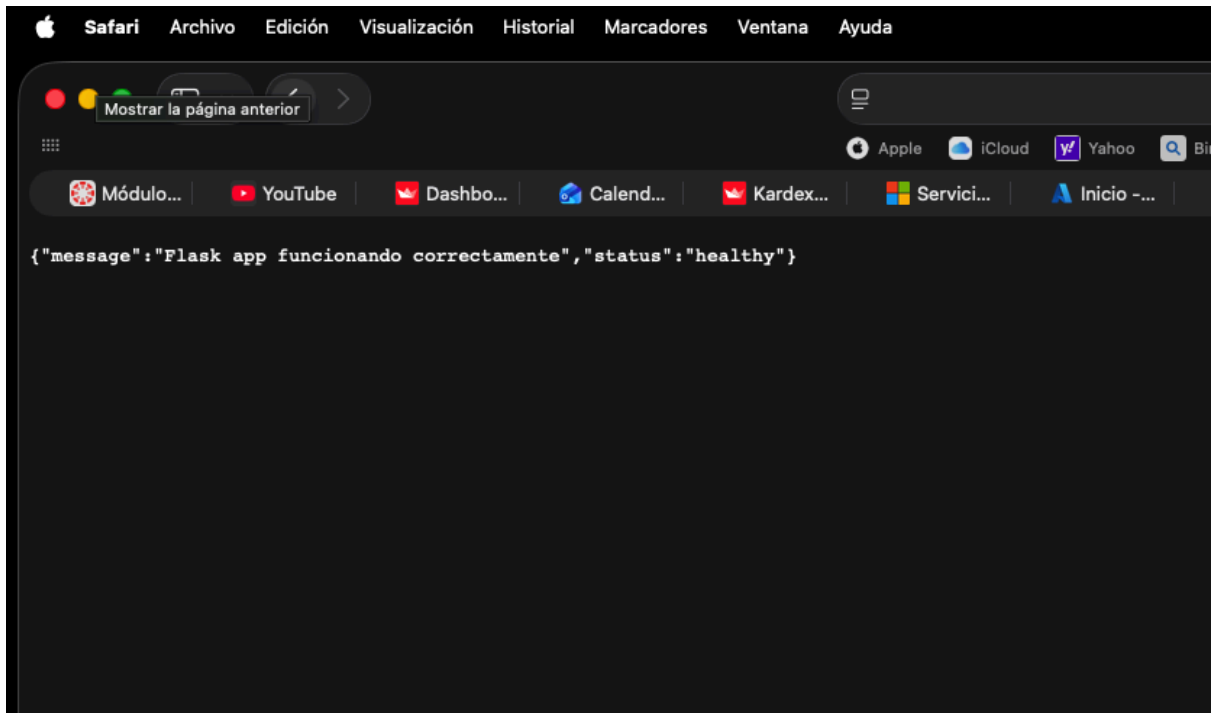
#### 4. Captura de su contenedor corriendo

```
flask-docker-app:1.0
c1bf5fc9b615161d731ea2a784631758580dfdb30ce4abff90dd8df038be4b7f
usuario@MacBookPro lab-docker-flask % docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS
c1bf5fc9b615   flask-docker-app:1.0   "gunicorn --bind 0.0..." 12 seconds ago Up 11 seconds 0.0.0.0
:8080->5000/tcp   mi-flask-app
usuario@MacBookPro lab-docker-flask %
```

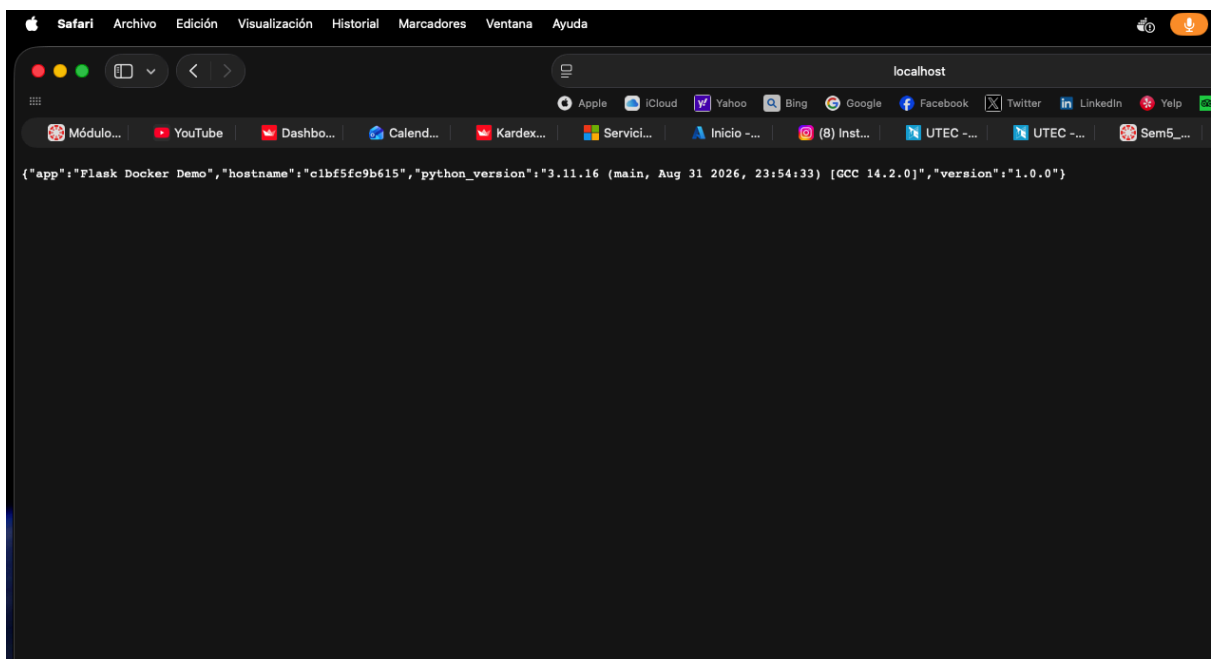
#### 5. Captura de su flask en el puerto 8080



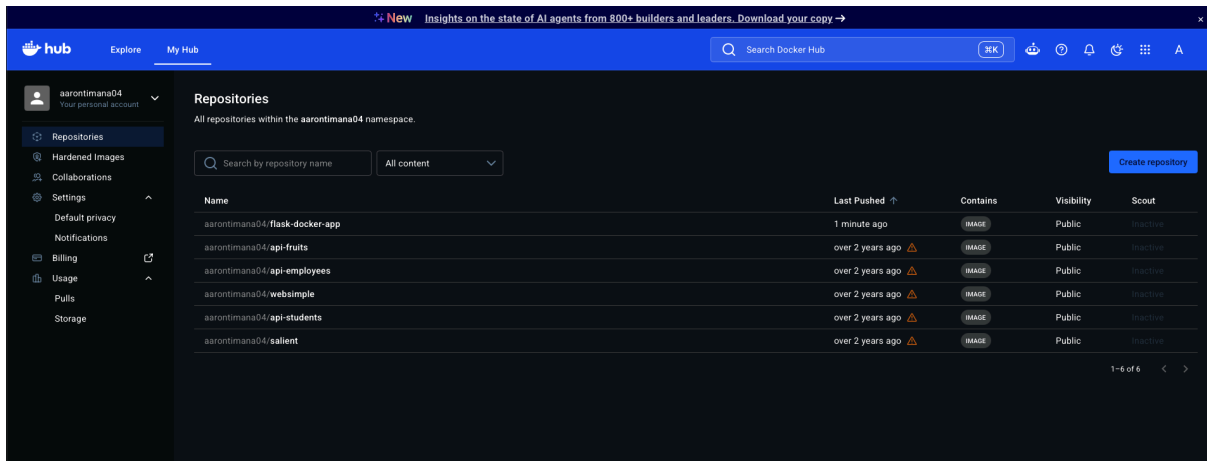
#### 6. Captura la api api/health



## 7. Captura la api api/info



## 8. Captura de su imagen publicada en docker hub



9. Reto: Cambiar la imagen base a python:3.11-alpine y comparar peso con la imagen python:3.11-slim

Captura con python:3.11-slim y python:3.11-alpine:

```
view build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/121qp0uqwx3szw0m1sc2k
p6m
[usuario@MacBookPro lab-docker-flask % docker images | grep flask-docker-app ]
flask-docker-app      alpine      ad180f7f570a  3 seconds ago  123MB
aarontimana04/flask-docker-app  1.0        31080e10a4dd  About an hour ago  249MB
aarontimana04/flask-docker-app  latest     31080e10a4dd  About an hour ago  249MB
flask-docker-app      1.0        31080e10a4dd  About an hour ago  249MB
usuario@MacBookPro lab-docker-flask % docker run -d \
--name mi-flask-alpine \
-p 8081:5000 \
[ flask-docker-app:alpine ]
25a3865cac6c666fe278759e3d5995d4c01a19537237c0080fae46760cf5723
[usuario@MacBookPro lab-docker-flask % docker ps ]
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
25a3865cac6c   flask-docker-app:alpine             "gunicorn --bind 0.0..." 7 seconds ago  Up 6 seconds
0.0.0.0:8081->5000/tcp   mi-flask-alpine
c1bf5fc9b615   abb66c23fbd7                       "gunicorn --bind 0.0..." About an hour ago  Up About an hour
0.0.0.0:8080->5000/tcp   mi-flask-app
[usuario@MacBookPro lab-docker-flask % docker images | grep flask-docker-app ]
flask-docker-app      alpine      ad180f7f570a  About a minute ago  123MB
flask-docker-app      1.0        31080e10a4dd  About an hour ago  249MB
aarontimana04/flask-docker-app  1.0        31080e10a4dd  About an hour ago  249MB
aarontimana04/flask-docker-app  latest     31080e10a4dd  About an hour ago  249MB
usuario@MacBookPro lab-docker-flask %
```

De 249MB a 123MB